

Dorian Turner

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Education

Imperial College London

Oct. 2024 – June 2027

BEng Computing

- **Degree:** 1:1 score achieved in Year 1. Predicted First Class (equiv. 4.0 GPA).
- **Relevant Modules:** Computer Systems - 82.53%, Databases - 78.04%, Algorithms - 77.33%.
- **Achievements:**
 - Elected as **year-wide well-being representative**, liaising between students and department
 - Most Interesting Extension, **ARMV8 C Project Prize** - Imperial College London
 - Encode ARC '26 **1st Place Overall** - best on-chain liquidity provider track
 - People's Choice Award** in Amazon University Engagement Program

Experience

Software Intern

London, England

Bloomberg

June 2026 - Ongoing

- Optimising the performance of a **latency-critical distributed microservice** in Bloomberg's Core Products division, responsible for validating data entitlements in the core data flow of the Bloomberg Terminal.
- Designing and implementing **request batching** in C++ to reduce network overhead, minimise redundant external service calls, and **improve tail latency** of the system.
- Analysing service bottlenecks through **profiling and performance testing** to identify inefficiencies and improve throughput in a high-scale distributed environment.
- **Collaborating with senior engineers** to evaluate system trade-offs, validate performance improvements, and ensure reliability and scalability across production workflows.

Software Intern

Oslo, Norway

Bridge Technologies

July 2025 – Aug. 2025

- Developed a machine learning based tool to derive MOS image scores for customers; it ran **15% faster** than the previous tool; was **32%** more correlated with human scores.
- Researched and developed a glitch detection algorithm to detect macro-blocking artifacts in live broadcast, to monitor thousands of streams.
- **Deployed LLMs** via a Slack chatbot for documentation Q&A and an FAQ website powered by a **local RAG** backend on a Mac Mini cluster.
- **Stack:** Python, OpenCV, FFmpeg, C++, ChatGPT, Local LLM, and FAISS.

Projects

FortDB (Independent Duo Project)

[Git](#) [🔗](#), July - Aug. 2025

- Developed a **thread-safe, NoSQL, append-only** database in C with another student.
- Designed **custom abstract syntax trees (ASTs)** for hierarchical data storage, enabling versioned path resolution, subdocument nesting, and time-travel queries with efficient traversal.
- Implemented **concurrency and memory primitives**: thread-safe hashmaps with versioned linked nodes.
- **Tools:** C POSIX standard library, C, Nix, Linux, and NixOS

Slice (Solo Project)

[Git](#) [🔗](#), Aug. 2026

- **Slice** is a Linux x86-64 **percentile profiler** that shows which **call-stack paths** occur during the slowest executions of one exact **C++ function**.
- Targets an exact **demangled ELF symbol** with **uprobes**, then combines **eBPF sampling**, invocation IDs, per-thread stacks, and percentile, thread, and time-range filters.
- On a **bimodally distributed C++ test service**, a **p95:p100 query isolates the tail** making the **slow function more prominent** than it is on the aggregate graph for the p00-p100 case.
- **Tools:** Rust, C++, DWARF, Linux, CMake, Ninja.

Technologies

Languages: C/C++, Java, Kotlin, Python, Haskell, Scala, Nix, SQL and Bash

Tools & Frameworks: Git, Docker, FFmpeg, OpenCV, FASTApi, Firebase, Claude Code, MCP and AI